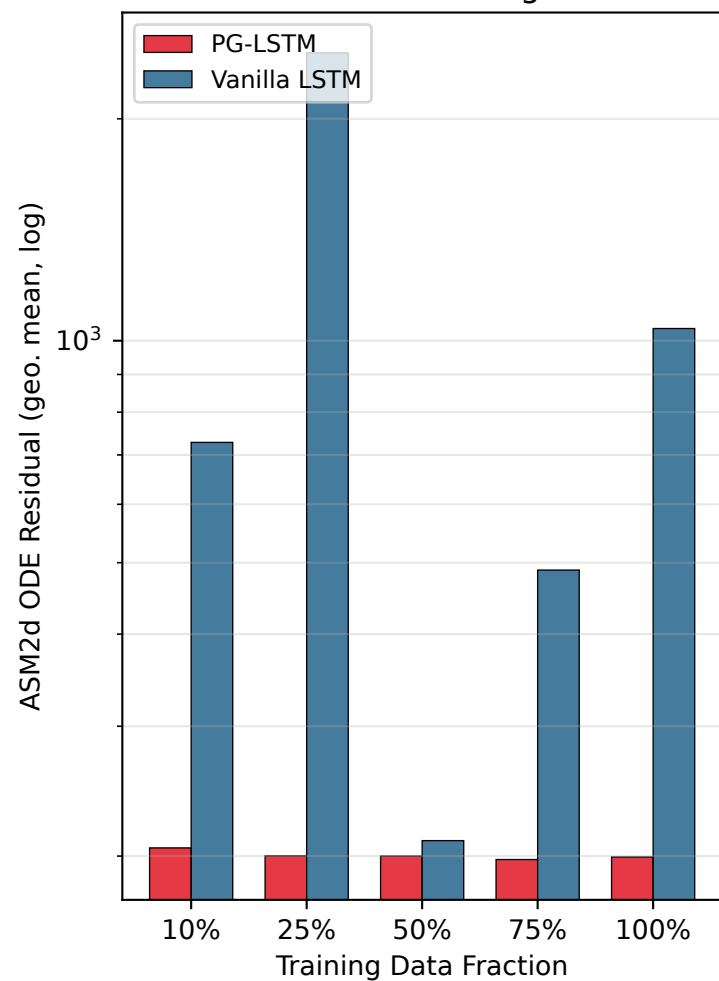
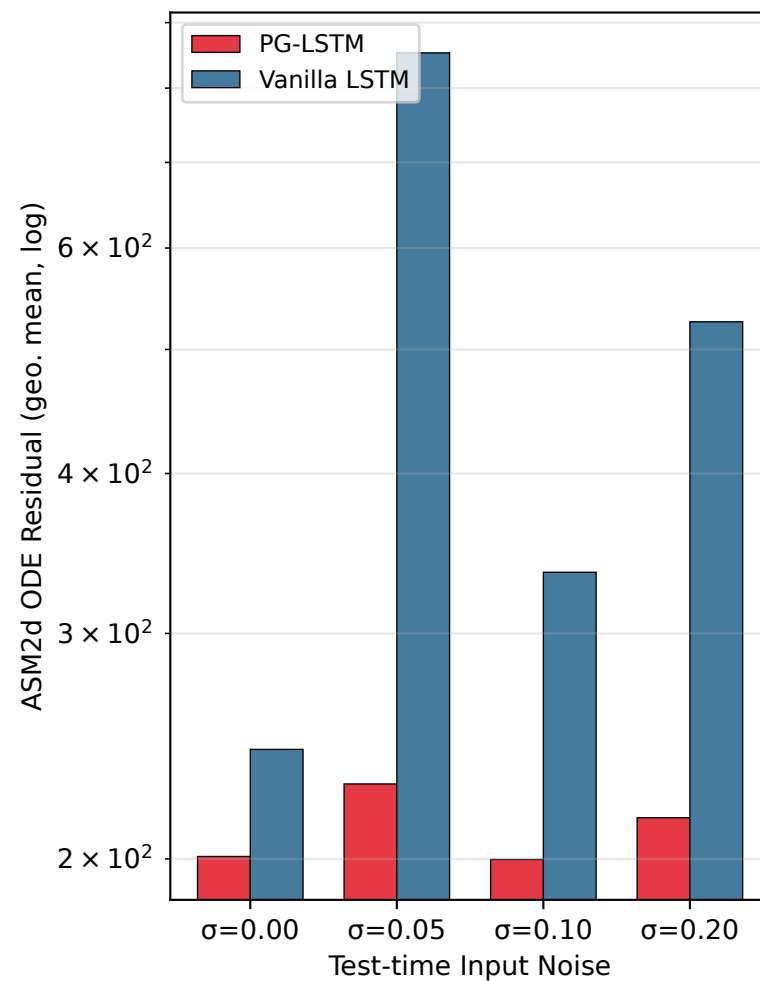


Domain of Utility for PG-LSTM in ASM2d Wastewater Modeling
Equivalent predictive accuracy (TOST $p < 0.05$) with a regime-dependent ASM2d ODE-residual advantage

A — Low-Data Regime



B — Noise Robustness



C — PG-LSTM Mechanistic Consistency Advantage Across Regimes

